English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Groups 4/5, 18.12.2021 Test 1

Your grade is 2 plus the number of points earned. Time allowed: 3 hours. Good luck.
Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (2 points) According to the data used in this problem, about 2% of the population has COVID-19. Test manufacturers report that rapid tests are positive with probability 65% for infected people and 1% for healthy people.

1. (0.5 points) A randomly selected person tests positive. What is the probability they are infected?
2. (0.5 points) In the setting of this problem, a negative test is needed for a green certificate. Suppose a person knows they are infected. How many tests on average are needed until a negative result[1]? What is the probability that two tests suffice to obtain the certificate?
3. (1 point) Consider two randomly selected people. What is the most likely pair (number infected, number of positive tests)? Calculate the correlation between these two counts, using approximations to accuracy 0.01.

Problem 2. (1.5 points) A casino offers this game: roll a die 100 times and receive BGN 1 for each six. The entry fee is BGN 20.

- (0.5 points) Let X_i be the winnings from rolls $25i-24$ through $25i$ for $i = 1, 2, 3, 4$, and let X be the total winnings. What are the distributions of these random variables? Would you play?
- (0.25 points) Let $Y = 4X_1$: instead of rolling 100 times, roll 25 times and multiply the winnings by 4. Compare the expectations and variances of X, Y, and briefly explain the result.
- (0.75 points) The casino offers an alternative game: if the sum of 100 rolls exceeds 555, you win BGN 1000. The entry fee is BGN 10. An advertisement calls this a special Christmas offer and claims the winning probability exceeds 1/100, giving positive expected profit. Can you confirm or refute this?

Problem 3. (0.5 points) Two points are chosen at random on a unit segment. Call its endpoints left and right. What is the probability that the three resulting segment lengths increase from left to right?

[1] For this problem, ignore the requirement to quarantine automatically after a positive result.